

Cross-validation, nested CV, bootstrap .632+

Workflow labs use the variant template: Goal → Approach → Execution → Check → Report.

Learning objectives

- Implement k-fold and repeated k-fold cross-validation from scratch and with a package, and report an honest generalisation error.
- Explain why selecting a model *and* tuning a hyperparameter on the same fold inflates performance, and use nested CV to fix it.
- Compute a bootstrap .632+ estimate and say when it is preferable to CV.

Prerequisites

Sampling variability; comfort with `caret` or base-R model fitting.

Background

Every predictive model reports an error on the data it was fit to. That number is almost always optimistic, because the fit has had a chance to memorise noise as well as signal. Cross-validation and the bootstrap are two resampling strategies that simulate the experience of scoring the model on data it has not seen. They are not interchangeable: they make different trade-offs between bias and variance, and they behave differently under high-dimensional settings.

A particular failure mode that recurs in the applied literature is *double dipping*: using the same held-out fold both to choose a hyperparameter and to report the final error. The fold is then no longer held out, and the reported error slides back toward the training error. Nested cross-validation is the clean solution: the inner loop picks hyperparameters, the outer loop reports error.

The bootstrap gives a different window onto generalisation. The plain bootstrap resamples with replacement; the .632 and .632+ variants weight the in-bag and out-of-bag errors to reduce bias when the learner overfits. In very small samples the .632+ estimator is often the most stable, and it is worth carrying in your toolbox alongside CV.

A practical note: when you write up a modelling study, the generalisation error is the headline number. If you tuned, you must disclose how. If the inner tuning cost was hidden in the outer fold, a careful reader will notice that the reported error is too good and cannot be reproduced. Nested CV is not a cosmetic fix; it is the difference between a number that generalises and a number that does not.

Setup

```
library(tidyverse)
library(glmnet)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Goal

Estimate the generalisation error of a lasso on a moderate p , small n simulated regression problem, first naively, then with a proper nested cross-validation.

2. Approach

We simulate $n = 120$ observations with $p = 40$ features, of which only 5 are true signal. We will (i) fit a single `cv.glmnet`, (ii) report the CV-minimising error, then (iii) wrap that in an outer 5-fold loop to show that the inner CV error is optimistic.

```
X <- matrix(rnorm(n * p), n, p)
beta <- c(rep(1.5, 5), rep(0, p - 5))
y <- as.numeric(X %*% beta + rnorm(n, sd = 1.0))

tibble(yhat_naive = as.numeric(X %*% beta), y = y) |>
  ggplot(aes(yhat_naive, y)) +
  geom_point(alpha = 0.6) +
  geom_abline(slope = 1, intercept = 0, colour = "grey50") +
  labs(x = "True linear predictor", y = "Observed y")
```

3. Execution

Naive single-layer CV.

```
naive_mse <- min(cv1$cvm)
naive_mse
```

Nested CV. The inner `cv.glmnet` picks λ ; the outer 5 folds produce the reported MSE.

```
folds <- sample(rep(1:K, length.out = n))
outer_mse <- numeric(K)
for (k in 1:K) {
  tr <- folds != k; te <- folds == k
  fit_k <- cv.glmnet(X[tr, ], y[tr], alpha = 1, nfolds = 5)
  yhat <- as.numeric(predict(fit_k, newx = X[te, ], s = "lambda.min"))
  outer_mse[k] <- mean((y[te] - yhat)^2)
}
nested_mse <- mean(outer_mse)
c(naive = naive_mse, nested = nested_mse)
```

4. Check

Bootstrap .632+ error as a cross-check.

```
errs_in <- errs_oob <- numeric(B)
for (b in 1:B) {
  idx <- sample.int(n, n, replace = TRUE)
  oob <- setdiff(seq_len(n), unique(idx))
  fit_b <- cv.glmnet(X[idx, ], y[idx], alpha = 1, nfolds = 5)
  yhat_in <- as.numeric(predict(fit_b, newx = X[idx, ], s = "lambda.min"))
  yhat_oob <- as.numeric(predict(fit_b, newx = X[oob, ], s = "lambda.min"))
  errs_in[b] <- mean((y[idx] - yhat_in)^2)
  errs_oob[b] <- mean((y[oob] - yhat_oob)^2)
}
err_app <- mean(errs_in); err_oob <- mean(errs_oob)
err_632 <- 0.368 * err_app + 0.632 * err_oob
```

```
c(apparent = err_app, oob = err_oob, boot632 = err_632,  
  nested = nested_mse)
```

5. Report

On simulated data ($n = 120$, $p = 40$, five true signals), a lasso with tuning parameter chosen by 5-fold CV achieved an out-of-sample MSE of `round(nested_mse, 2)` under nested 5×5 CV. The naive single-layer CV understated the error at `round(naive_mse, 2)`; the bootstrap .632 estimator gave `round(err_632, 2)`, which bracketed the nested estimate.

The naive number looks tighter, but it reflects the choice of `lambda` as well as the fit. Only the nested number is a fair estimate of what this procedure — tune-then-fit — will do on new data.

Common pitfalls

- Tuning on the test fold and reporting the resulting error.
- Using a single 80/20 split in high dimensions; variance across splits can exceed the effect of interest.
- Bootstrapping without tracking in-bag vs out-of-bag, which inflates the estimate back to the apparent error.

Further reading

- Hastie, Tibshirani, Friedman, *Elements of Statistical Learning*, §7.10–7.11.
- Varma S, Simon R (2006), *Bias in error estimation when using cross-validation for model selection*.
- Efron B, Tibshirani R (1997), *Improvements on cross-validation: the .632+ bootstrap method*.

Session info

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)